

Oracle Cloud Capacity Analytics Extracts User's Guide

Oracle Cloud Capacity Analytics
Enterprise Manager Cloud Control (EMCC) Extract
Automatic Workload Repository (AWR) Miner Extract

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Introduction

This Sizing Extracts User's Guide is intended to assist customers in running a Data Collection from the Oracle Enterprise Manager Cloud Control (EMCC) or from the Database's Automatic Workload Repository (AWR) for purposes of database system sizing and/or database assessment.

The EMCC Extract process documented here must run against the EMCC Repository Database that is monitoring the Target set of databases. The extract process itself can run from any host or desktop that has SQL*Plus installed and Oracle Net (SQL*Net) access to the Repository Database.

The AWR Miner Extract process documented here must run against each Target database that must be Oracle Database Enterprise Edition. The extract process itself can run from any host or desktop that has SQL*Plus installed and Oracle Net (SQL*Net) access to the Target Database.

License Exception

Both Extract tools rely on tables and views that belong to the Oracle Diagnostics Pack. Per MOS [Doc ID 2771065.1](#), an exception is granted for the purpose of running these scripts in order to collect sizing data.

Tools and Version Support

EMCC Extracts data collection support list:

- EMCC 13.x and later supported
- **EMCC 12.x and older not supported**

The version of EM Agents installed on each target is EXTREMELY IMPORTANT because this impacts the data collected by the agent for that target. Older agent versions do not collect important metrics used by OCCA and may require overrides (entered in Properties files).

AWR Miner data collection support list:

- Oracle Database Enterprise Edition 11g and later supported
- **Oracle Database Enterprise Edition 10g and older not supported**
- **Oracle Database Standard Edition any version not supported**

The version of AWR Miner should be 25.2.0 or higher to be supported with OCCA. Older OCCA versions or versions for other sources such as BAR or AWS stand of good chance of failing with current versions of OCCA, ensure you are on a current version of the script.

Note: *Both tools support Multitenant Databases, but only the metrics collected from the CDB root can be used for sizing at this time, i.e., multiple pluggable databases within a CDB cannot be independently sized.*

EMCC Extract Process Overview

1. Select Host or Desktop with SQL*Plus and access to EMCC Repository Database. The space required is depended on the size of the EMCC, typically you will need 50 to a few 100 Megabytes, but for very large installations it can be several Gigabytes. The platform can be most any Unix flavor (Linux, Solaris, AIX, HP-UX) or Windows (requires PowerShell enabled)
2. Verify Username (Account), Password and SQL*Net Connect String to the EMCC Repository Database. If you use the SYSMAN account then access to the objects will all be setup, if you choose to use a different account be sure it has been properly setup by following MOS [Doc ID 2583605.1](#). You can verify that your access is setup using the following SQL statement:

```
SQL> select count(*) from sysman.mgmt$target;  
COUNT(*)  
-----  
880
```

If the count returned is not greater than zero, then there is an issue that must be addressed before proceeding.

3. Run the [Get_EMCC_Sizing_Interactive.\[sh|ps1\]](#) script to perform the EMCC Extract, syntax and examples are listed below.

Note: The Windows PowerShell execution policy may need to be adjusted, see [Appendix D](#) for more information on this.

4. **Optional:** The Extract script will pull data for every database in the repository and Oracle will filter it to look at only the in-scope databases, however, if you have a need to filter the output yourself you can this by running a Python script that is outlined in [EMCC Extract - Source Side Filtering](#) appendix below.
5. **Optional:** The Extract script will pull list database names and host names if this creates an issue, security or otherwise, you do have the ability to obfuscate the output before it is sent back to Oracle by running a Python script that is outlined in the [EMCC Extract - Source Side Obfuscation](#) appendix below.
6. After any optional post processing is completed, you can zip up the files and send them back to Oracle, typically your Oracle contact will have setup a SharePoint folder and shared it with you so that you can upload the files to that location, as the files even zipped are often too large for email.

AWR Miner Extract Process Overview

1. Select Host or Desktop with SQL*Plus and access Target database. The space required is depended on the target database, typically you will need a few to maybe 100 Megabytes per database. The platform can be most any Unix flavor (Linux, Solaris, AIX, HP-UX) or Windows.
2. Verify Username (Account), Password and SQL*Net Connect String to the Target Database. If you are using a local connection and connecting as SYSDBA then access will not be an issue, if using a different account make sure the following minimal privileges have been granted to the username:

```
grant create session to USERNAME
grant select_catalog_role to USERNAME
```

3. Run the SQL*Plus script like any other script, for example:

```
sqlplus USERNAME/PASSWORD@HOST.ACME.COM/ORADB.ACME.COM @awr_miner.sql
```

Note: *If this is a RAC database the script only needs to be run against one of the instances as the DBA_HIST_ views contain data from all instances.*

4. Script output is automatically saved to a file in the current working directory with a name in the following format, please do not rename the files as it will break the processing on the Oracle side.

```
awr-hist-<DBID>-<DBNAME>-<BEGIN-SNAP>-<END-SNAP>.out
```

5. **Optional:** The AWR Miner script will include database names and host names in the output files, if this creates an issue, security or otherwise, you do have the ability to obfuscate the output before it is sent back to Oracle by running a Python script that is outlined in the [AWR Miner - Source Side Obfuscation](#) appendix below.
6. After any optional post processing is completed, you can zip up the files and send them back to Oracle, typically your Oracle contact will have setup a SharePoint folder and shared it with you so that you can upload the files to that location, as the files even zipped are often too large for email.

EMCC Extract Steps and Examples

Step 1: Unzip the Customer Extract Package

Download or move the zip file provided by your Oracle contact, to the OS with the Oracle client installed and unzip it.

The zip file will create a sub-directory in your current location called extracts, and the extracted files will be placed in this location. Below is an example of what you can expect see from the unzip command:

```
$ unzip /tmp/Customer_emcc_sizing_extracts.zip
Archive:  /tmp/Customer_emcc_sizing_extracts.zip
  creating: extracts/
  inflating: extracts/Get_EMCC_Sizing_Host_Props.sql
  inflating: extracts/Get_EMCC_Sizing_Structure_DB.sql
  inflating: extracts/Get_EMCC_Sizing_DB_Params.sql
  -- snipped output --
  inflating: extracts/Get_EMCC_Sizing_Structure_VM.sql
  inflating: extracts/Get_EMCC_Sizing_DB_Props.sql
  inflating: extracts/Get_EMCC_Sizing_Metrics_Host_daily_03_months.sql
  inflating: extracts/Get_EMCC_Sizing_Run_Hourly.sh
```

Step 2: Validate Prerequisites - SQL*Plus and EM Connect String

The Enterprise Manager Cloud Control extract scripts rely on a SQL*Plus feature that was introduced in version 12.2, so they will not work correctly with any older version.

Warning: You must use **SQL*Plus client version 12.2 or higher** for this process to work. The EM Repository Database can be an older version, it is just the client that needs to be at the higher version.

The scripts can be run from the EM Repository Database host or from a client that is installed on a different host if Oracle Net (aka SQL*Net) can be used to access the database. A driver script is provided automate the running of the SQL scripts. On Linux/Unix this driver script is a BASH & KSH shell compatible script, and on Windows the driver script is a PowerShell script.

Whether the collection is performed from a Windows or Linux client, 2 pieces of information will be required. The Path to the supported version of SQL*Plus, and the Oracle Net Connection string with username and password. Below are examples for collecting and validating these 2 pieces of information.

Retrieve Path to valid SQL*Plus command:

```
$ whereis sqlplus
sqlplus: /u01/app/oracle/product/19.0.0/dbhome_1/bin/sqlplus

$ /u01/app/oracle/product/19.0.0/dbhome_1/bin/sqlplus -v
SQL*Plus: Release 19.0.0.0.0 - Production
Version 19.13.0.0.0
```

Retrieve Connect String to connect to the EM Repository Database:

```
$ sqlplus sysman/private1@emhost.acme.com/emdb.acme.com
SQL*Plus: Release 19.0.0.0.0 - Production on Mon Oct 24 17:07:06 2022
Version 19.13.0.0.0
Copyright (c) 1982, 2021, Oracle. All rights reserved.
Last Successful login time: Mon Oct 24 2022 16:18:34 -05:00
Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.5.0.0.0
```

Step 3: Run EMCC Collection for Each EM Repository

The collection script needs 2 things to run successfully, both items were verified in Step 2 so you can use them here.

1. SQL*Plus command that meets the version requirement
2. Connect String with username and password to access the Enterprise Manager Repository Database

You can optionally enter one or both pieces of information on the command line. If Connect String isn't provided on the command line the script will prompt for it, as it has in the example below. If a path for SQL*Plus not provided on the command line the script will search for a suitable SQL*Plus command in your environment, then in the ORATAB, and finally, if needed, it will prompt you for the path.

Note: The run time for the script is dependent multiple factors. Typically, the expected run time would be in the 15-45 minutes range but on very large environments it could run for a few hours. On Unix/Linux the collection script can be run using the nohup command if both optional command line arguments are provided when it is executed.

Script Syntax:

Linux Command Line: :

```
./Get_EMCC_Sizing_Interactive.sh [CONNECT_STRING] [SQLPLUS_PATH]
nohup ./Get_EMCC_Sizing_Interactive.sh CONNECT_STRING SQLPLUS_PATH >log.out 2>&1 &
```

Windows Command Prompt:

```
powershell .\Get_EMCC_Sizing_Interactive.ps1 [CONNECT_STRING] [SQLPLUS_PATH]
```

Windows PowerShell Prompt:

```
.\Get_EMCC_Sizing_Interactive.ps1 [CONNECT_STRING] [SQLPLUS_PATH]
```

Where:

CONNECT_STRING	would look something like the following: sysman/private@emhost.acme.com:1527/emdb.acme.com	
SQLPLUS_PATH	would look something like one of the following: /u01/app/oracle/product/19.0.0/dbhome_1 /oracle/instantclient_19_1 /u01/app/oracle/product/19.0.0/dbhome_1/bin/sqlplus	(ORACLE_HOME) (PATH_TO_DIR) (FULL_PATH)

Note: If **SQLPLUS_PATH** argument is not provided on the command line, then the script will check the environment and/or ORATAB file for the path, or if that fails the script prompts for the required information.

Example Executions:

Linux Example, Arguments provided on Command Line

```
/home/oracle/extracts $ ./Get_EMCC_Sizing_Interactive.sh
emread/private1@sysman/private1@emhost.acme.com/emdb.acme.com
/u01/app/oracle/product/19.0.0/dbhome_1

This script queries views licensed by the Oracle Diagnostics Pack;
however, the license is not required per MOS Note: 2771065.1

SUCCESS: SQL*Plus Command Found and Version is good:
"/u01/app/oracle/product/19.0.0/dbhome_1/bin/sqlplus"
SUCCESS: Test Connection to Database Succeeded!
SUCCESS: Test Access to EM View Succeeded!
SUCCESS: Test Access to EM View Found Rows!

Starting Extract Scripts!

Extracting DB HA relationships...
-- snipped output --
Extracting PDB hourly metrics 0 to 03 months...

Checking Output Files for Errors!

Run Time = 6 minutes

SUCCESS: No Errors were found in the output files!

Next Step: bundle up the output directory, for example:
zip -r emcc_sizing_extracts_oem1.zip emcc_sizing_extracts

Final Step: please upload the bundle to location provided by your Oracle contact
```

Windows Example, Script Prompts for Arguments

```
C:\Users\oracle\extracts>powershell .\Get_EMCC_Sizing_Interactive.ps1

This script queries views licensed by the Oracle Diagnostics Pack;
however, the license is not required per MOS Note: 2771065.1

SQL*Plus version 12.2 or higher is required!
Enter Path to SQL*Plus Command: c:\oracle\instantclient_12_2
SUCCESS: SQL*Plus Command Found and Version is good: c:\oracle\instantclient_12_2\sqlplus.exe

Enter Enterprise Manager Repository Database Connection Data
The following are examples for the syntaxes you may use
EZCONNECT: username/password@hostname:port/service_name
TNSNAMES : username/password@tnsnames_alias

Note: The User must have at least read-only access, or have SYSMAN privileges

Enter Connection Data: emread/private1@sysman/private1@emhost.acme.com/emdb.acme.com
SUCCESS: Test Connection to Database Succeeded!
SUCCESS: Output Directory Created: .\emcc_sizing_extracts

Starting Extract Scripts!

Extracting DB HA relationships...
-- snipped output --
Extracting PDB hourly metrics 0 to 03 months...

Checking Output Files for Errors!

SUCCESS: No Errors were found in the output files!

Next Step: bundle up the output directory, for example:
zip -r emcc_sizing_extracts_oem1.zip emcc_sizing_extracts

Final Step: please upload the bundle to location provided by your Oracle contact
```

Step 3a: Optional - Filter the Extracted Data

You do not need to filter the data before sending it to Oracle, in fact the process is improved if you send all the extracted data to Oracle. The data you send to Oracle as part of sizing estimate or other analysis, is only ever shared to the select group of Oracle employees that are working on your effort. It is not shared with other groups inside of Oracle.

Sharing all of the data will make it possible to perform a mini review on the health and hygiene of your EM environment. Often, we can offer suggestions and recommendations that can improve your EM solution.

If you decide you do want to filter the data before it is sent to Oracle, please go to the [EMCC Extract - Source Side Filtering](#) appendix below and follow the steps there.

Step 3b: Optional - Obfuscate the Extracted Data

Same as above you do not need to obfuscate the data before sending it to Oracle, and again like above not obfuscating the data can help in many ways.

in fact, the process is improved, if you send all the extracted data to Oracle. The data you send to Oracle as part of sizing estimate or other analysis, is only ever shared to the select group of Oracle employees that are working on your effort. It is not shared with other groups inside of Oracle.

Sharing all of the data will make it possible to perform a mini review on the health and hygiene of your EM environment. Often, we can offer suggestions and recommendations that can improve your EM solution.

If you decide you do want to filter the data before it is sent to Oracle, please go to the [EMCC Extract - Source Side Filtering](#) appendix below and follow the steps there.

Step 4: Create a Zip file and Upload the Extracts

Following the successful execution of the steps above bundle up the extracted metrics, note that depending on the Optional steps the directory which you want to zip up may be one of the following:

```
emcc_sizing_extracts
emcc_sizing_extracts_filtered
emcc_sizing_extracts_obfuscated
```

Zip up the correct directory:

```
$ zip -r emcc_sizing_extracts.zip emcc_sizing_extracts
adding: emcc_sizing_extracts/ (stored 0%)
adding: emcc_sizing_extracts/emcc_sizing_structure_db_params.csv (deflated 93%)
-- snipped output --
adding: emcc_sizing_extracts/emcc_sizing_metrics_db_daily_12_months.txt (deflated 34%)
adding: emcc_sizing_extracts/emcc_sizing_metrics_pdb_daily_06_months.csv (deflated 97%)
```

Finally, after the files have been bundled up using zip, upload the file to the location provided by your Oracle contact, and let your Oracle contact know it's been uploaded, and you're done.

AWR Miner Extract Steps and Examples

Select Host or Desktop with SQL*Plus and access to EMCC Repository Database, space required is

Step 1: Unzip the Customer Extract Package

Download or move the zip file provided by your Oracle contact, to the OS with the Oracle client installed and unzip it. The package should be named like one of the following:

- Customer_emcc_sizing_extracts.zip *Includes both EMCC and AWR Miner Extract scripts and Guides*
- AWR_Miner.zip *Includes only AWR Miner Extract scripts and Guides*

Step 2: Validate Prerequisites

In order to run the SQL*Plus script you will need a connection to the database, and you will need a database account with the required privileges.

Database Connection: The connection to the database can be a local connection such as "/ as sysdba" or a remote connection using SQL*Net. If the target database is Multitenant, then connection needs to be to the CDB root container.

Hint: *Using remote connections can allow for simple automation of running this tool against many databases in a very short time.*

Database Account: The database account will need at least the following minimal privileges:

```
create session
select_catalog_role
```

Step 3: Run AWR Miner Extract Collection for Each Database

Below are a couple of examples for running the script. In both cases you can see that the script automatically generates an output file with a unique name, so the script can be run from a single location against many databases.

Example Executions:

```
Local Connection, no SQL*Net, connect as sysdba

[oracle@labhost1 ~]$ sqlplus / as sysdba @awr_miner.sql
SQL*Plus: Release 19.0.0.0.0 - Production on Tue Oct 28 10:46:57 2025
Version 19.25.0.0.0
-- snipped output --
AWR-Miner Version 25.2.0
Collects metrics and statistics from AWR History of individual Oracle Databases.
```

```
-- snipped output --
Starting Extract Script!
Extracting OS information...
Extracting DB patch History...
Extracting Module...
Extracting DB AWR snapshots...
-- snipped output --
Extracting SQL statements by snapshot...
Extracting Row counts...
Completed Extract Script!
Collection File : awr-hist-1532163575-C19A-6660-6866.out
-----
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.25.0.0.0
```

Remote Connection, SQL*Net and Credentials

```
[oracle@labhost2 ~]$ sqlplus c##readuser/XXXX@labhost1.acme.com/c19a.acme.com @awr_miner.sql
SQL*Plus: Release 23.0.0.0.0 - for Oracle Cloud and Engineered Systems on Tue Oct 28 11:20:54
2025
Version 23.6.0.24.10
-- snipped output --
AWR-Miner Version 25.2.0
Collects metrics and statistics from AWR History of individual Oracle Databases.
-- snipped output --
Starting Extract Script!
Extracting OS information...
Extracting DB patch History...
Extracting Module...
Extracting DB AWR snapshots...
-- snipped output --
Extracting SQL statements by snapshot...
Extracting Row counts...
Completed Extract Script!
Collection File : awr-hist-1532163575-C19A-6660-6867.out
-----
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.25.0.0.0
```

Step 3b: Optional - Obfuscate the Extracted Data

You do not need to obfuscate the data before sending it to Oracle, in fact the process is improved if obfuscation is not used. The data you send to Oracle as part of sizing estimate or other analysis, is only ever shared to the select group of Oracle employees that are working on your effort. It is not shared with other groups inside of Oracle.

If you decide you do want to obfuscate the data before it is sent to Oracle, please go to the [AWR Miner - Source Side Obfuscation](#) appendix below and follow the steps there.

Step 4: Create a Zip file and Upload the Extracts

Following the successful execution for each of the databases, collect all of the `awr-hist-*.out` files to a single location and bundle them up, note that depending on the Optional steps the directory name may have changed.

Note: Do not rename out files, the Oracle processing tool will expect the original file names for all of the `awr-hist-*.out`.

Zip up the correct directory:

```
$ zip awr_miner_out.zip awr-hist-*.out
adding: awr-hist-1379172541-DHSEEP-65452-65652.out (172 bytes security) (deflated 90%)
adding: awr-hist-2109698798-NSP1CDB-46500-46701.out (172 bytes security) (deflated 91%)
-- snipped output --
adding: awr-hist-2385585065-DSPPROD-62286-62492.out (172 bytes security) (deflated 90%)
adding: awr-hist-4040750112-ADP1CDB-28425-28629.out (172 bytes security) (deflated 90%)
```

```
$ ls -lhd awr_miner_out.zip
-rw-r--r-- 1 user 197121 1.5M Sep 26 10:56 awr_miner_out.zip
```

Finally, after the files have been bundled up, upload the file(s) to the location provided by your Oracle contact, and let your Oracle contact know it's been uploaded, and you're done.

Appendix A: EMCC Extract - Source Side Filtering

Source Side Filtering adds the ability to filter the Enterprise Manager data before it is sent to Oracle for review and processing.

It runs as a post processing step following the normal extract covered above and creates a new set of files containing data for only those targets that have not been selected for exclusion in the filter file.

Note: If you plan to also run the Data Obfuscation step listed below in Appendix B, you should run this Filtering script first and then you will need to rename the output directories so that Obfuscate script finds the Filtered files.

Prerequisites

- Desktop or host with Python3 (3.6 or higher) installed. Python2 (2.7.18 or higher) may continue to function but using a current version of Python3 is recommended.
- All Extracted files from the EMCC Repository, from Step 3 listed above.

Run the Filtering Script

1. Open a terminal window and change directory to the `extracts` directory that contains both the `emcc_sizing_extracts` directory and the `Filter_Extracts.py` file.
2. Verify the Python installation, on some systems multiple versions of Python may exist, use the higher version if both exist:

```
[oracle@emhost extracts]$ cat /etc/oracle-release
Oracle Linux Server release 9.6

[oracle@emhost extracts]$ python3 -V
Python 3.9.21
```

3. At the command prompt run the Filter script with the "`--create-filter`" option and the latest Python version, for example:

```
[oracle@emhost extracts]$ python3 Filter_Extracts.py --create-filter
Created directory :/home/oracle/extracts/emcc_sizing_extracts_filtered
=====
Filter_Extracts.py, version 1.6
=====
Wrote 18 lines to
/home/oracle/extracts/emcc_sizing_extracts_filtered/properties_database_filter.csv
```

4. Using a spreadsheet, such as Excel or LibreOffice Calc, edit the `properties_database_filter.csv` file highlighted above, and add the Yes to column A (Exclude) for those databases you do not want to be included in the filtered output.

	A	B	C	D	E	F	G	H
1	Exclude	Database Name	dbmachine	dbmachine_ow	cluster	Server List	cdb_version	cdb_status
2	Yes	oecent	DB Machine exa5.acme.com	SYSMAN	Cluster-c1	exa5-03vm01.acme.com,exa5-02vm01.acme.com,exa5-04vm01.acme.com	19.7.0.0.0	Pending/Unknown
3	Yes	oewest	DB Machine exa5.acme.com	SYSMAN	Cluster-c1	exa5-03vm01.acme.com,exa5-02vm01.acme.com,exa5-04vm01.acme.com	19.7.0.0.0	Target Down
4		swboe	DB Machine exa5.acme.com	SYSMAN	Cluster-c1	exa5-03vm01.acme.com,exa5-02vm01.acme.com,exa5-04vm01.acme.com	19.7.0.0.0	Target Up
5	Yes	dbsg	DB Machine exa5.acme.com	SYSMAN	Cluster-c2	exa5-03vm02.acme.com,exa5-02vm02.acme.com,exa5-04vm02.acme.com	19.7.0.0.0	Target Down
6		DM	DB Machine exa5.acme.com	SYSMAN	Cluster-c2	exa5-03vm02.acme.com,exa5-02vm02.acme.com,exa5-04vm02.acme.com	19.7.0.0.0	Target Up
7		rtl1	DB Machine exa5.acme.com	SYSMAN	Cluster-c2	exa5-03vm02.acme.com,exa5-02vm02.acme.com,exa5-04vm02.acme.com	19.7.0.0.0	Target Up
8		dclone2	DB Machine exa5.acme.com	SYSMAN	Cluster-c3	exa5-04vm03.acme.com,exa5-03vm03.acme.com,exa5-05vm03.acme.com	19.10.0.0.0	Target Up
9		dclone3	DB Machine exa5.acme.com	SYSMAN	Cluster-c3	exa5-04vm03.acme.com,exa5-03vm03.acme.com,exa5-05vm03.acme.com	19.10.0.0.0	Target Up
10	Yes	demodb	DB Machine exa5.acme.com	SYSMAN	Cluster-c3	exa5-04vm03.acme.com,exa5-03vm03.acme.com,exa5-05vm03.acme.com	19.10.0.0.0	Target Down
11	Yes	SOEDEV	DB Machine exa5.acme.com	SYSMAN	Cluster-c3	exa5-04vm03.acme.com,exa5-03vm03.acme.com,exa5-05vm03.acme.com	19.10.0.0.0	Target Down
12	Yes	testdev	DB Machine exa5.acme.com	SYSMAN	Cluster-c3	exa5-04vm03.acme.com,exa5-03vm03.acme.com,exa5-05vm03.acme.com	19.10.0.0.0	Target Down
13	Yes	TM	DB Machine exa5.acme.com	SYSMAN	Cluster-c3	exa5-04vm03.acme.com,exa5-03vm03.acme.com,exa5-05vm03.acme.com	19.10.0.0.0	Target Down
14	Yes	prim	DB Machine exa5.acme.com	SYSMAN	Cluster-c4	exa5-04vm04.acme.com,exa5-03vm04.acme.com,exa5-05vm04.acme.com	19.7.0.0.0	Target Down
15		em13c0			none	em13c0.cloud.osc.oracle.com	19.5.0.0.0	Target Up

- After the file has been updated and returned to its location in the `emcc_sizing_extracts_filtered` directory run the Filter script again with the "--filter" option, for example:

```
[oracle@emhost extracts]$ python3 Filter_Extracts.py --filter
=====
Filter_Extracts.py, version 1.6
=====
Filtering files...

Targets : Filtered[11] Total[18]
emcc_sizing_availability_history.csv [#####] 100% [2097*] [0.002]s
-- snipped output --
emcc_sizing_structure_host_vm.csv [#####] 100% [0] [0.001]s

Filtering completed in [5.926]s
```

Note: Following the filter operation and before you zip up the filtered data, please check to see that the `properties_database_filter.csv` file no longer exists within the `emcc_sizing_extracts_filtered` directory. If the file does exist there, please move it to a different location. That file contains unfiltered data which you may not want to send to Oracle.

- If you don't plan to Obfuscate the output, then go to [Step 4](#) above, but remember the directory to zip up is now: `emcc_sizing_extracts_filtered`

-- or --

If you are planning to Obfuscate the output, proceed to the [EMCC Extract - Source Side Obfuscation](#) appendix.

Appendix B: EMCC Extract - Source Side Obfuscation

Source Side Obfuscation can be used when server names, database names, owners, etc. can't be shared with Oracle.

Note: Data extracted from any EMCC Repository will often have some issues, missing data, agents down, etc., these issues are often harder to resolve once the data is obfuscated, so more time and coordination with Oracle will often be required.

Prerequisites

- Desktop or host with Python3 (3.6 or higher) installed. Python2 (2.7.18 or higher) may continue to function but using a current version of Python3 is recommended.
- All Extracted files from the EMCC Repository, from Step 3 listed above or from the Source Side Filtering in Appendix A.

Run the Obfuscation Script

1. Open a terminal window and change directory to the `extracts` directory that contains both the `emcc_sizing_extracts` directory and the `Obfuscate_Extracts.py` file.
2. Verify the Python installation, on some systems multiple versions of Python may exist, use the higher version if both exist:

```
[oracle@emhost extracts]$ cat /etc/oracle-release
Oracle Linux Server release 9.6

[oracle@emhost extracts]$ python3 -V
Python 3.9.21
```

3. At the command prompt run the Obfuscate script with the latest Python version, for example:

```
[oracle@emhost extracts]$ python3 Obfuscate_Extracts.py
=====
Obfuscate_Extracts.py, version 1.7
=====
Begin @ 2022-10-25 16:16:46 UTC
Obfuscate_Extracts.py, version 1.7
sys.version_info : sys.version_info(major=3, minor=10, micro=6, releaselevel='final',
serial=0)
platform.system() : Linux
=====
DIR          : ./emcc_sizing_extracts
APPEND0      : False
DEBUG        : False
PROCESS_TELEMETRY: True
ROOT_DIR     : .
=====
Obfuscating ./emcc_sizing_extracts/emcc_sizing_structure_db.csv...
Wrote 15 rows to file ./emcc_sizing_extracts_obfuscated/emcc_sizing_structure_db.csv
-- snipped output --
Obfuscating ./emcc_sizing_extracts/emcc_sizing_metrics_host_daily_03_months.csv...
Wrote 7032 rows to file
    ./emcc_sizing_extracts_obfuscated/emcc_sizing_metrics_host_daily_03_months.csv

Wrote translation to file: ./emcc_sizing_output/obfuscated_translation.csv

All files written to directories rooted @ extracts
Processing completed in 3.775 s
End @ 2022-10-25 16:16:49 UTC
```

4. The obfuscated files are written to a new directory named `emcc_sizing_obfuscated`. A new file will also be created that translates the original values to the obfuscated values, it is written to the `obfuscated_translation.csv` file in the `emcc_sizing_output` directory. Make sure and keep a copy of this file as it may be required to answer questions from Oracle.
5. Return to [Step 4](#) above to bundle up and upload the files.

Appendix C: AWR Miner - Source Side Obfuscation

Source Side Obfuscation can be used when server names and database names can't be shared with Oracle.

Prerequisites

- Desktop or host with Python3 (3.6 or higher) installed. Python2 (2.7.18 or higher) may continue to function but using a current version of Python3 is recommended.
- Collect all the AWR Miner output files, from [Step 3](#) above into a directory named `awr_miner_out` in the `extracts` directory

Run the Obfuscation Script

1. Open a terminal window and change directory to the `extracts` directory that contains both the `awr_miner_out` directory and the `Obfuscate_Awr.py` file.
2. Verify the Python installation, on some systems multiple versions of Python may exist, use the higher version if both exist:

```
[oracle@host extracts]$ cat /etc/oracle-release
Oracle Linux Server release 9.6

[oracle@host extracts]$ python3 -V
Python 3.9.21
```

3. At the command prompt run the Obfuscate script with the latest Python version, for example:

```
[oracle@host extracts]$ python3 Obfuscate_Awr.py
=====
Obfuscate_Awr.py, version 1.4
sys.version_info : sys.version_info(major=3, minor=9, micro=21, releaselevel='final',
serial=0)
platform.system() : Linux
=====
Created directory :/home/oracle/extracts/awr_miner_out_obfuscated
Using existing directory :/home/oracle/extracts/awr_miner_occa/log

Obfuscating /home/oracle/extracts/awr_miner_out/awr-hist-106797285-EBSPROD-9839-
10052.out....
Wrote 18900 rows to file /home/oracle/extracts/awr_miner_out_obfuscated/awr-hist-
106797285-oracle-database-00000-9839-10052.out....
-- snipped output --
Obfuscating /home/oracle/extracts/awr_miner_out/awr-hist-551143314-EBSTST-33437-
33639.out....
Wrote 29062 rows to file /home/oracle/extracts/awr_miner_out_obfuscated/awr-hist-
551143314-oracle-database-00005-33437-33639.out....

Wrote translation to file:
/home/oracle/extracts/awr_miner_occa/awr_obfuscated_translation.csv
```

4. The obfuscated files are written to a new directory named `awr_miner_out_obfuscated`. A new file will also be created that translates the original values to the obfuscated values, it is written to the `awr_obfuscated_translation.csv` file in the `awr_miner_out` directory. Make sure and keep a copy of this file as it may be required to answer questions from Oracle.
5. Return to [Step 4](#) above to bundle up and upload the files.

Appendix D: Enable Windows PowerShell Execution

The Windows PowerShell execution policy is set up by default as Restricted and doesn't allow the scripts to run, however if you have administrator privileges this can be changed.

Output below shows the type of error you will see if PowerShell is not allowed to execute:

```
C:\> powershell .\Get_EMCC_Sizing_Interactive.ps1
.\Get_EMCC_Sizing_Interactive.ps1 : File C:\Get_EMCC_Sizing_Interactive.ps1
cannot be loaded because running scripts is disabled on this system.
For more information, see about_Execution_Policies at
https://go.microsoft.com/fwlink/?LinkID=135170.
At line:1 char:1
+ .\Get_EMCC_Sizing_Interactive.ps1
+ ~~~~~
+ CategoryInfo          : SecurityError: (:) [], PSSecurityException
+ FullyQualifiedErrorId : UnauthorizedAccess
```

You can also check if PowerShell scripts are allowed to run by opening a command window and running the following:

```
C:\> powershell Get-ExecutionPolicy
Restricted
```

If, as in the case above, it comes back with **Restricted**, the execution policy will not allow PowerShell Scripts to run.

To enable the PowerShell Scripts, open a new command window using the "Run as administrator" option, and run the following command:

```
C:\> powershell Set-ExecutionPolicy RemoteSigned

C:\> powershell Get-ExecutionPolicy
RemoteSigned
```

Note: If your Security requires it, you can disable the PowerShell privilege after running the tool by setting the value back to **Restricted**.

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